

Write legibly showing all the steps VERY clearly. Max. score=100 (Total points = 105).

1 Define the following:

Let Ω be a non-empty set, $\mathcal{F}$ is a collection of subsets of Ω and $P : \mathcal{F} \rightarrow \mathbb{R}$ is a function.

- (1) $(\Omega, \mathcal{F}, P)$ is called a probability space if ...
- (2) A function X , defined on a probability space $(\Omega, \mathcal{F}, P)$ is called a random variable (r.v) if ...

Let X be a random variable defined on $(\Omega, \mathcal{F}, P)$, with c.d.f F_X and law $P_X \equiv \mu_{F_X}$.

- (3) X is a *Exponential*(λ) random variable ($\lambda > 0$ is a constant) if ...
- (4) X is called a *singular* variable if ...

Let $\{X_n\}$, X be random variables defined on some $(\Omega, \mathcal{F}, P)$.

(5) $\{X_n\}$ is a sequence of independent and identically distributed (i.i.d.) random variables if ...

- (6) X_n converges to X "almost surely (a.s)" or "with probability 1 (w.p. 1)" or "a.e (P)" if ...

- (7) X_n converges to X in probability (w.r.t P) if ...

- (8) if each $X_n \in L^1(\Omega, \mathcal{F}, P)$, then the collection $\{X_n\}$ is called uniformly integrable (UI) if ...

20

(3×8 = 24 points)

2 Let X, Y be two independent random variables on some probability space $(\Omega, \mathcal{F}, P)$, with marginal laws P_X, P_Y on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$. Recall that X and Y independent iff their joint law is the *product measure* of their marginal laws, i.e $P_{(X,Y)} = (P_X) \times (P_Y)$ on $(\mathbb{R}^2, \mathcal{B}(\mathbb{R}^2))$. Let m is the Lebesgue measure on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ and $m^2 \doteq m \times m$ is the Lebesgue measure on $(\mathbb{R}^2, \mathcal{B}(\mathbb{R}^2))$. Let f, g be two non-negative functions in $L^1(\mathbb{R}, \mathcal{B}(\mathbb{R}), m)$.

- (a) If X and Y are independent and $E(X^2) < \infty$, $E(Y^2) < \infty$, then carefully prove

$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y).$$

- (b) Suppose $P_X \ll m, P_Y \ll m$, with $\frac{dP_X}{dm} = f$ and $\frac{dP_Y}{dm} = g$. Then prove

$$P_{(X,Y)} \ll m^2, \quad \frac{dP_{(X,Y)}}{dm^2} = f \cdot g \quad \text{a.e } (m^2)$$

— something with Tonelli

$$f_n \rightarrow f$$

- (c) If f is continuous and bounded on $\mathbb{R}$, so is $f * g$.

$$|f(x-n)/(g(n))| < m g(n)$$

- (d) If $\lambda \doteq P_X * P_Y$ Prove $\frac{d\lambda}{dm} = f * g$ a.e (m), f and g are as in (b).

by continuous

✓ (e) If Z is a random variable with $P_Z = \lambda$ as in (d) above, then how is Z related to X and Y ? What does the result in (d) say about the law of Z ? bounded

(6+6+6+6+(3+3)= 30 points)

Z is the sum of X and Y

the law of Z is the distribution of $(X+Y)$

Let $\{X_n\}_{n \geq 1}$ be a sequence of *nonnegative* random variables defined on a probability space $(\Omega, \mathcal{F}, P)$. Let

$$Y(\omega) = \liminf_{n \rightarrow \infty} X_n(\omega), \quad \omega \in \Omega.$$

should be

able

10

- (i) Prove $Y : \Omega \rightarrow \mathbb{R}$ is a random variable.
 (ii) If $\limsup_{n \rightarrow \infty} E(X_n) < \infty$, then prove that $Y \in L^1(\Omega, \mathcal{F}, P)$.
 (iii) If $M(t)$ is the moment generating function of Y , i.e. $M(t) = E(e^{tY})$, $t \in \mathbb{R}$, then show

$$P(Y > a) \leq \inf_{t > 0} e^{-ta} M(t), \quad a > 0.$$

Marlov

(5+5+5= 15 points)

similar

6

- 4 Let m be the Lebesgue measure on $[0, 1]$ and $I_A(x)$ denote the indicator function of A . Consider the following sequence of random variables on $(\Omega = [0, 1], \mathcal{F} = \mathcal{B}[0, 1], P = m)$: For $n \geq 1$, $X_n(\omega) = nI_{(0, \frac{1}{n^2})}(\omega)$ and $X(\omega) = 0$ for $\omega \in \Omega$.

- (i) Show $X_n \rightarrow X$ almost surely (P).
 (ii) Does $X_n \rightarrow X$ in probability?
 (iii) Show $X_n \rightarrow X$ in $L^p \equiv L^p(\Omega, \mathcal{F}, P)$ for $0 < p < 2$, but this convergence does not hold for $2 \leq p < \infty$.
 (iv) Show that $X_n \rightarrow X$ in L^p for some $0 < p < \infty$ implies that $X_n \rightarrow X$ in L^q , for any $0 < q < p$.

Holder's

(3+3+(3+2)+5 = 16 points)

- 5 Let $\alpha \in \mathbb{R}, \beta > 0$ be constants.

- (i) Show that a $Normal(\alpha, \beta^2)$ random variable exists:

In other words, using steps of the Caratheodory extension theorem, exhibit a probability space $(\Omega, \mathcal{F}, P)$ and a random variable $X : \Omega \rightarrow \mathbb{R}$, such that its c.d.f. is given by Φ where

$$\Phi(x) = \int_{-\infty}^x \frac{1}{\sqrt{2\pi}\beta^2} e^{-\frac{(u-\alpha)^2}{2\beta^2}} du, \quad x \in \mathbb{R}.$$

- (ii) Compute the following Lebesgue integral (Clearly stating any result used)

$$\int_{\Omega} X(\omega) dP(\omega).$$

Komogorov consistency theorem

- (iii) For each $n \geq 1$, let μ^n be a probability measure on $(\mathbb{R}^n, \mathcal{B}(\mathbb{R}^n))$. Under what conditions can we say that there is a probability measure P on $(\mathbb{R}^\infty, \mathcal{B}(\mathbb{R}^\infty))$ such that $P(A \times \mathbb{R}^\infty) = \mu^n(A)$ for any $A \in \mathcal{B}(\mathbb{R}^n)$, for any $n \geq 1$? Describe a sequence of random variables $\{X_n : n \geq 1\}$ defined on $(\Omega = \mathbb{R}^\infty, \mathcal{F} = \mathcal{B}(\mathbb{R}^\infty), P)$ such that the joint distribution of $(X_1, \dots, X_n)$ is given by μ^n , for all $n \geq 1$.

- (iv) Using part (iii) or otherwise, show that there exists $\{X_n : n \geq 1\}$ which is an i.i.d. sequence of $Normal(\alpha, \beta^2)$ random variables.

(5+5+5+5= 20 points)

Let $f(\alpha) = \Phi(x)$
 or $\mu = \Phi(x)$